

Test Cases and Results

Total Test Cases	Passed	Failed
7	7	0

Test Summary

Test ID	Test Description	Expected Result	Actual Result	Status
TC01	Successful booking	Booking becomes CONFIRMED and a voucher is generated.	Payment successful: INR 6000.0; booking confirmed; voucher V-TC01-B displayed.	PASS
TC02	Overlapping booking rejected	Booking rejected because the room is unavailable.	Booking rejected: Room is not available.	PASS
TC03	Payment failure	Booking is not confirmed and no voucher is generated.	Payment failed; booking NOT confirmed; no voucher generated.	PASS
TC04	Cancellation — full refund	Booking is cancelled and refund equals the total booking amount.	Refund percentage 100.0%; refund INR 16000.0.	PASS
TC05	Cancellation — partial/no refund	Refund is less than total according to the cancellation policy.	14 hours before check-in; refund percentage 0.0%; refund INR 0.0.	PASS
TC06	Non-overlapping dates	Both bookings are allowed, including a second booking starting at checkout.	Both bookings confirmed and vouchers displayed.	PASS
TC07	Invalid room	Room lookup fails and the invalid room is rejected.	findRoom(999) returned null.	PASS

Detailed Test Cases

TC01 — Successful booking

Objective: Booking becomes CONFIRMED and a voucher is generated.

Input / Setup: H001 / room 101; 2030-01-10 to 2030-01-12; payment=true.

Expected Result: Booking becomes CONFIRMED and a voucher is generated.

Actual Result: Payment successful: INR 6000.0; booking confirmed; voucher V-TC01-B displayed.

Status: PASS

TC02 — Overlapping booking rejected

Objective: Booking rejected because the room is unavailable.

Input / Setup: H001 / room 101; 2030-01-11 to 2030-01-13, overlapping TC01.

Expected Result: Booking rejected because the room is unavailable.

Actual Result: Booking rejected: Room is not available.

Status: PASS

TC03 — Payment failure

Objective: Booking is not confirmed and no voucher is generated.

Input / Setup: H001 / room 102; 2030-02-10 to 2030-02-12; payment=false.

Expected Result: Booking is not confirmed and no voucher is generated.

Actual Result: Payment failed; booking NOT confirmed; no voucher generated.

Status: PASS

TC04 — Cancellation — full refund

Objective: Booking is cancelled and refund equals the total booking amount.

Input / Setup: H001 / room 103; 2030-03-10 to 2030-03-12; cancellation 2030-03-01T10:00.

Expected Result: Booking is cancelled and refund equals the total booking amount.

Actual Result: Refund percentage 100.0%; refund INR 16000.0.

Status: PASS

TC05 — Cancellation — partial/no refund

Objective: Refund is less than total according to the cancellation policy.

Input / Setup: H002 / room 201; 2030-04-10 to 2030-04-12; cancellation 2030-04-09T10:00.

Expected Result: Refund is less than total according to the cancellation policy.

Actual Result: 14 hours before check-in; refund percentage 0.0%; refund INR 0.0.

Status: PASS

TC06 — Non-overlapping dates

Objective: Both bookings are allowed, including a second booking starting at checkout.

Input / Setup: H002 / room 202; first 2030-05-10 to 2030-05-12; second 2030-05-12 to 2030-05-14.

Expected Result: Both bookings are allowed, including a second booking starting at checkout.

Actual Result: Both bookings confirmed and vouchers displayed.

Status: PASS

TC07 — Invalid room

Objective: Room lookup fails and the invalid room is rejected.

Input / Setup: H001 / room 999.

Expected Result: Room lookup fails and the invalid room is rejected.

Actual Result: findRoom(999) returned null.

Status: PASS

Execution Result

All seven required test cases completed successfully: 7 PASS and 0 FAIL.